

PART-I

6) Following are the step and ramp responses of the closed-loop system with PI Controller constructed using pole placement technique, with maximum percent overshoot is smaller than %4 and settling time is smaller than 10 seconds.

It can be seen that the system can track the step response with zero steady-state error. Nevertheless it can track the ramp input with a steady-state error. This can be explained due to the concept of system types, where the system in the question is Type-1, which can track a step input with zero steady-state error and a ramp input with a non-zero steady-state error. As the system has only one free integrator due to Integral control action, the system is Type-1.

